INDEPENDENT NOVELTY AUDIT · PUBLICATION FEASIBILITY

Novelty Audit: Detection-Conditioned Cell Insertion for Table Structure Recognition

A claim-by-claim literature verification of the proposal “Learning to Insert: Image-Grounded Cell Recovery from Incomplete Table Structure Predictions” — covering the 2023–2026 TSR literature, reference verification, closest prior art, reviewer threat analysis, and an ICDAR 2027 publication plan.

Prepared for the proposing author

Scope: broad-sweep novelty search · verification of all cited 2025–2026 references ·
venue and timeline analysis for a solo, single-GPU research setup

TARGET VENUE: ICDAR 2027 · KUALA LUMPUR

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VERDICT SNAPSHOT

1. Executive Summary

SURVIVES	10 / 10 REAL	28 FEB 2027
Core novelty claim holds after a broad 2023-2026 sweep - no direct prior found	Cited 2025-2026 references verified; only two minor metadata quirks	ICDAR 2027 listed submission deadline - about 4.3 months of runway

The proposal under review, **Learning to Insert: Image-Grounded Cell Recovery from Incomplete Table Structure Predictions**, defines a deliberately narrow intervention in table structure recognition (TSR). Instead of building another end-to-end table parser, it proposes a recovery stage that observes a table image together with the surviving cell candidates of a fixed base detector, and learns to insert additional cell instances only where image evidence and surviving-cell context justify them. The insertion decoder may output a variable number of new candidates, may explicitly decline to insert, and may recognize that an apparent vacancy belongs to an existing spanning cell. The author asks whether this design is novel enough for ICDAR or a Q1 journal, and this report answers that question with a claim-by-claim audit against the 2023-2026 literature.

The audit verified every 2025-2026 reference that decides the novelty question, then swept the TSR literature broadly: a curated community repository covering more than 150 methods from 2017 to 2026, ICDAR 2024/2025/2026 proceedings coverage, the 2026 arXiv output (TableSeq, FastTab, StrucTab, ConRTF, InstructTable), the historical-record reconstruction line in IJDAR, and the adjacent domains that a reviewer would raise - missed-detection recovery in object detection, semi-supervised pseudo-label completion, two-stage DETR query generation, and the vision-language-model (VLM) table parsing wave. Search phrasing covered missing-cell recovery, candidate insertion, cell hypothesis generation, detection completion, and vacancy detection variants.

Headline finding: the narrow contribution survives. No published TSR system was found that learns a variable-cardinality insertion operator conditioned on an incomplete, detector-produced candidate set. What exists today is rule-based geometric gap filling (the IJDAR 2025 historical-record work infers missing cells from the alignment of detected cells), structure-recovery heuristics (LGPMA searches and merges empty cells on its own segmentation), span merging from a complete grid (Formerge), and dynamic query generation inside primary detectors (DQ-DETR-style TSR, DINO-style two-stage query selection). None of these conditions a learned decoder on surviving candidates, and none implements an explicit insert / no-insert / span-compatible decision. The three novelty pillars - detection-conditioned learned insertion, the three-way insertion decision, and the controlled-deletion plus natural-omission evaluation package - therefore stand, provided the paper is written against the closest prior art identified in Chapter 5.

Three risks remain, none of them novelty-fatal. First, the empirical risk: on the clean, canonicalized PubTables-1M benchmark, a strong geometric completion baseline or simple threshold relaxation could capture many natural omissions, which would shrink the headline gain; the go/no-go gate in the proposal

already anticipates this. Second, positioning risk: VLM-based table parsers are now published at top venues (StrucTab at ECCV 2026; UniTable reports surpassing GPT-4o on table parsing), so the paper must argue explicitly why a detector-level intervention still matters. Third, timing: ICDAR 2027 convenes in Kuala Lumpur in August 2027 and its listed paper deadline is 28 February 2027, roughly 4.3 months away rather than the six months assumed - the plan in Chapter 7 compresses accordingly.

Recommendation: proceed. The proposal is unusually well calibrated - its own novelty-boundary table survives re-assessment almost unchanged - and the reference list is real. Keep the host detector frozen, keep the architecture small, run the two baselines first (threshold relaxation and geometric completion), and let the natural-omission experiment decide whether this becomes a full architecture paper or a careful negative-result analysis. Both outcomes are publishable at ICDAR if measured honestly; only the headline framing would change.

WHAT WAS SEARCHED AND HOW

2. Audit Scope and Method

The audit had two stages. The first stage verified the existence and content of the ten 2025-2026 references that the proposal itself flags as decisive, plus one dataset reference (PubTables-v2). Verification meant locating the paper on arXiv, Springer, ScienceDirect, or a proceedings site; confirming the author list, venue, and year; and, for the closest priors, reading the abstract or full text to check that the proposal characterizes them accurately. This step matters because AI-assisted drafts sometimes cite plausible but nonexistent work, and any such fabrication would have invalidated the proposal's novelty boundary.

The second stage was a broad novelty sweep anchored on the task definition - learned, variable-cardinality insertion of cell candidates into an incomplete detector output. Primary sources were: a curated table-structure-recognition repository maintained by a community researcher, which indexes roughly 150 methods from 2017 to 2026 with venues; ICDAR 2024 and 2025 proceedings entries (SepFormer, TABLET, DTSM, ETT, MuTabNet, SPRINT, ClusterTabNet, and peers); ICDAR 2026 acceptances visible on arXiv (ConRTF); 2026 arXiv preprints (TableSeq, FastTab, StrucTab, InstructTable); and the image-to-sequence and VLM line (TableFormer, VAST, UniTable, TabPedia, InstructTable, StrucTab). Adjacent domains were swept as well: missed-detection and false-negative recovery in object detection, semi-supervised object detection with pseudo-label completion (the Soft Teacher line), two-stage DETR and DINO query selection, and DETR-variance analysis relevant to query design.

Query phrasings included combinations of: missing cell recovery, missed cell detection, cell completion, candidate insertion, cell hypothesis, detection completion, vacancy detection, empty cell recovery, threshold relaxation, and proposal omissions, crossed with table structure recognition and with generic object detection. The IJDAR 2025 historical-record paper was fetched in full to verify the exact mechanism of its missing-cell inference, because it is the closest direct prior and the proposal's novelty argument leans on distinguishing learned insertion from that rule-based completion.

Three limitations apply. The sweep relied on web-indexed sources, so paywalled full texts other than the closest priors were checked at abstract level only. Non-English venues were not systematically covered,

although the dominant TSR publication venues are English-language. And any literature search has a tail latency: work published in the weeks before submission will not appear here. The proposal already contains the correct mitigation - re-run a focused search immediately before submission and adjust the novelty claim if needed. This report's Chapter 8 reading list is designed to make that re-verification a one-hour task rather than a fresh research project.

DO THE DECISIVE CITATIONS EXIST

3. Reference Verification (2025-2026)

The proposal cites a cluster of very recent work (2025-2026) that defines the state of the art it must be measured against. Each entry below was located and cross-checked against the proposal's description. The result is reassuring: every cited item exists, and the descriptions match the sources with only two minor metadata discrepancies, noted beneath the table.

Ref	Proposal claim	Verification	Notes
[15] TableCenterNet	One-stage parallel regression of spatial and logical cell locations (arXiv:2504.17522)	VERIFIED	arXiv Apr 2025; public GitHub (dreamy-xay); matches claim.
[16] Historical-record TSR	IJDAR 2025; infers missing cells from alignment of detected positive cells	VERIFIED	Singh and Middleton, IJDAR vol. 28 (s10032-025-00543-9); mechanism quote confirmed verbatim; already accumulating citations.
[17] Semi-supervised historical TSR	IJDAR 2025; evaluates missing-cell generation in a semi-supervised setting	VERIFIED	Singh et al., IJDAR (s10032-025-00562-6), published online Dec 2025.
[21] OG-HFYOLO	Deformation table cell instance segmentation (arXiv:2504.20682)	VERIFIED	Title differs slightly: actual title uses Orientation Gradient Guidance and Heterogeneous Feature Fusion; fix wording when citing.
[22] ConRTF	Edge-constrained boundary refinement for realtime transformer TSR (arXiv:2607.00734)	VERIFIED	Accepted to ICDAR 2026 per arXiv comments; D-FINE based; up to +1.6 GriTS at equal inference speed.
[23] TableSeq	Unified autoregressive structure, content, and coordinates (arXiv:2604.16070)	VERIFIED	Apr 2026; image-only single-stream model; strong PubTabNet/FinTabNet/SciTSR results.
[24] FastTab	Tiny recursive module plus 1D transformers (arXiv:2605.22422)	VERIFIED	May 2026; grid-centric low-latency TSR.
[25] StrucTab	Structured optimization with intermediate supervision and decomposed RL rewards (arXiv:2606.29905)	VERIFIED	Now an ECCV 2026 paper with public code and the TableVerse-5K benchmark - a notable VLM-side advance the proposal should cite as such.

Ref	Proposal claim	Verification	Notes
[26] BGTR	Bounding-box-guided HTML generation (arXiv:2609.08705)	VERIFIED with dating quirk	The work is ICPR 2024; the arXiv posting (Sep 2026) is archival. Cite as ICPR 2024, not 2026.
[27] TPGCAT	Text prior-guided cell alignment, Pattern Recognition 114832	VERIFIED	In press / available Sep 2026 (S0031320326017966).
PubTables-v2	Optional extension dataset (multi-page)	VERIFIED	Smock et al., arXiv:2512.10888 (Dec 2025); first large-scale multi-page TSR benchmark; also a useful future testbed.

Table 3.1 - Verification status of the 2025-2026 references that decide the novelty question.

Two metadata corrections are required before submission. Reference [21] should quote the actual OG-HFYOLO title (Orientation Gradient Guidance and Heterogeneous Feature Fusion for Deformation Table Cell Instance Segmentation), and reference [26] should be dated ICPR 2024 since the September 2026 arXiv entry is an archival upload of already published work. Neither affects the novelty argument.

One bonus finding outside the requested verification window: reference [19] cites TEDS as Evaluating Table Structure Recognition: A New Perspective (arXiv:2208.00385). The TEDS metric originates from Zhong, Xie, and Liu, Image-based Table Recognition: Data, Model, and Evaluation (ECCV 2020, arXiv:1905.13391), which introduced the tree-edit distance similarity and its structural variant S-TEDS. The proposal should cite that source, or both, to avoid a reviewer flagging a mis-attributed metric. Separately, two verified status upgrades strengthen the related-work map rather than weakening it: StrucTab is now peer-reviewed at ECCV 2026, and ConRTF is accepted at ICDAR 2026, which confirms that both the VLM/RL line and the detector-refinement line remain active through the submission window.

THE CORE ASSESSMENT

4. Novelty Audit: Claim-by-Claim Verdicts

The proposal's own Table 1 divides candidate claims into established territory and a narrow surviving contribution. This chapter re-assesses that boundary with the evidence gathered in the sweep. The overall conclusion is that the proposal's self-assessment is accurate and, if anything, slightly conservative: the intervention it proposes was not found anywhere in the 2017-2026 TSR literature, while everything it concedes as established is indeed established.

Claim	Audit verdict	Strongest evidence
Recover missing cells from surviving detections (alignment or grid based)	ESTABLISHED	IJDAR 2025 historical-record work generates candidate cells at alignment gaps (mechanism verified verbatim); RobusTabNet-style split-and-merge pipelines imply grid completion.
Find vacancies using row/column alignment or topology	ESTABLISHED	LGPMA empty-cell searching and merging; historical-record gap filling; split-and-merge grid vacancy logic.

Claim	Audit verdict	Strongest evidence
Use image evidence together with structural information for recovery	ESTABLISHED	LGPMA couples local/global visual mask alignment with structural recovery; modern joint spatial-logical predictors (LORE, TGRNet, TableCenterNet) blend both sources.
Refine cell geometry with structural context	ESTABLISHED	Boundary refinement is an active detector-side line (ConRTF, ICDAR 2026; VAST visual alignment).
Create new structural cells inside a reconstruction pipeline	ESTABLISHED	TableFormer PDF-oriented orphan handling; historical-record reconstruction creates cells at gaps.
Learned insertion of omitted instances from an incomplete detector set, with variable output cardinality	NARROW-NOVEL - the core claim	No prior found across curated 2017-2026 TSR listings, ICDAR 2024-2026 proceedings coverage, 2026 arXiv, and adjacent-domain sweeps. Closest mechanism relatives are primary detectors with dynamic queries (DQ-DETR TSR, DINO two-stage query selection), which do not condition on a surviving candidate set.
Explicit insert / no-insert / span-compatible three-way decision	NARROW-NOVEL - differentiating	Formerge merges grid cells into spans (cardinality reduction); LGPMA divides empty cells by rules; no learned per-vacancy three-way decision exists. DETR's background class is per-query objectness in a primary detector, not a span-compatibility outcome.
Controlled-deletion training plus filtered-vs-proposal omission evaluation package	NARROW-NOVEL - measurement contribution	Deletion-style robustness tests exist generically in detection, but the assembled protocol (filtered vs proposal omissions, insertion precision and recall, false insertion rate, span violation rate, no-change accuracy) was not found as a task-level evaluation in TSR.

Table 4.1 - Re-assessed novelty boundary. The proposal's own assessment survives.

4.1 Pillar 1 - learned, variable-cardinality insertion

The search found no system that takes an incomplete candidate set produced by a base detector and emits a variable number of new cell instances through a learned decoder conditioned on that set. The rule-based completion in the IJDAR 2025 historical-record paper is the nearest functional neighbor: it generates candidate cells where gaps are identified from the horizontal and vertical alignment of detected positive cells, but it is a fixed geometric heuristic with no learned insertion decision, no image-grounded scoring at insertion time, and no ability to abstain. LGPMA recovers empty cells, yet it does so inside its own mask-alignment segmentation pipeline rather than on top of an external detector's candidate set, and its recovery is a search-and-merge procedure. Query-based detectors such as TATR and DQ-DETR-style TSR generate cell hypotheses from image content, but as primary detectors - they never observe a surviving candidate set, and their per-query background class is not a decision about a specific geometric vacancy in a given incomplete table. This is the gap the proposal fills, and it is a genuine one.

4.2 Pillar 2 - the three-way insertion decision

The insert / no-insert / span-compatible outcome space is the most differentiating design element. Spanning-cell handling exists, but always as a separate operation: split-and-merge methods merge grid elements into spans (SPLERGE, SEM, Formerge), and LGPMA locates and divides empty cells with rules. No learned operator was found that decides, per vacancy, whether a new independent cell, no cell, or a span-compatible explanation is correct. The closest conceptual relative is the background class of a DETR query, which teaches a query to output nothing, but that is objectness in a primary detector rather than a structural explanation of a vacancy conditioned on neighboring cells. If the experiments show this three-way head reduces over-segmentation of spanning cells relative to a box-only ablation, it becomes the paper's most defensible architectural contribution.

4.3 Pillar 3 - the evaluation package

The controlled-deletion protocol with isolated, adjacent, and span-adjacent deletion patterns; the split of natural omissions into filtered omissions (a candidate existed below the operating threshold) versus proposal omissions (no candidate existed even in a generous pre-filter pool); and the insertion-specific metrics (insertion precision, recovery recall, false insertion rate, candidate recall gain, span violation rate, intact-table no-change accuracy) together form an evaluation contribution that the sweep did not find elsewhere in TSR. Robustness-by-deletion appears in detection generally, and competition reports often analyze missed detections, but the proposal's package is assembled into a task-level protocol with matched-precision comparison against threshold relaxation and geometric completion. As a standalone contribution it would be thin; as the experimental backbone of the learned-insertion paper it is exactly what a skeptical reviewer needs.

4.4 Hypotheses H1-H5 under audit

Hypothesis	Status	What the paper needs
H1 - candidate incompleteness is a non-trivial error source	PLAUSIBLE, UNMEASURED	Phase 1 profiling on PubTables-1M with TATR: report the natural omission rate at the operating threshold and the filtered/proposal split. Caveat: PubTables-1M is canonicalized and clean, so omissions may be rarer than in wild scans - state rates explicitly.
H2 - threshold relaxation is not equivalent to learned insertion	MUST-BE-TESTED (baseline)	Matched-precision comparison at equal candidate budgets; report where lowered thresholds create duplicates or competing false candidates.
H3 - learned insertion recovers natural proposal omissions	CORE EMPIRICAL CLAIM	Recovery recall on proposal omissions at matched insertion precision; this single number decides the paper's fate.
H4 - both image evidence and surviving-cell context matter	ABLATION-SUPPORTED DESIGN	The 2x2 information-source ablation (no-recovery control, context-only, image-only, full model) is well designed; keep it.
H5 - explicit no-insertion reduces false insertions and span damage	TESTABLE VIA ABLATION	With-vs-without no-insertion class ablation; report span violation rate and intact-table no-change accuracy.

Table 4.2 - Hypothesis status and the evidence each requires.

DOSSIERS ON THE EIGHT NEAREST NEIGHBORS

5. Closest Prior Art: Deep Analysis

This chapter gives each methodological neighbor a short dossier: what it does, what it leaves open, and what the future paper must do to differentiate itself. Together these dossiers define the related-work section of the submission and the baselines a reviewer will expect.

5.1 LGPMA - empty-cell searching and merging (ICDAR 2021)

LGPMA aligns cell regions with local and global pyramid masks and then runs a uniform structure recovery pipeline of cell matching, empty cell searching, and empty cell merging, which locates and splits empty cells so that both non-empty and empty cells are handled. It won the ICDAR 2021 best industry paper award and remains the standard citation for structure-aware vacancy handling. What it leaves open: recovery happens inside its own segmentation pipeline with rule-driven searching and merging, not as a learned operator over an external detector's candidate set; there is no explicit abstain outcome and no notion of returning no-insertion on an intact table. The paper must present LGPMA as the strongest established combination of image evidence and topological vacancy reasoning, and position insertion as a different intervention point: after a fixed detector, on its retained candidate set.

5.2 Singh and Middleton (IJDAR 2025) - alignment-based missing-cell inference

This is the closest direct prior, and the audit verified its mechanism from the full text. The authors state the procedure explicitly:

> *Since the trained model generates a limited number of table cells (up to 2000, including both positive and negative predictions), we implemented a method to infer missing cells based on the horizontal and vertical alignment of detected positive cells. This approach ensures complete table reconstruction by generating candidate cells where gaps are identified and aligning them with existing cells.*
>
> *Singh and Middleton, IJDAR vol. 28, 2025, Section 3.2*

The method is exactly the coordinate-based geometric completion that the proposal demotes to baseline status: no learned insertion decision, no image evidence at insertion time, no abstention class, and a historical-records domain with CascadeTabNet as the detector. It establishes that generating candidates at detected gaps is not new, which the proposal already concedes. For the submission, this paper is simultaneously a threat and an asset: it is the geometric-completion baseline to reimplement on the same detector, and it proves the field recognizes missing-cell reconstruction as a real problem worth solving.

5.3 Singh et al. (IJDAR 2025) - semi-supervised historical TSR

The companion paper studies semi-supervised table structure recognition for digitizing historical tables and evaluates missing-cell generation and reconstruction within that setting. Its relevance is contextual rather than competitive: it shows the historical-table community actively works on incomplete structure recovery, and it supplies a domain where omissions are frequent and where a recovery module could later be stress-tested. It leaves the supervised, detection-conditioned intervention study untouched, and it does not constrain the novelty claim.

5.4 Formerge - recovering spanning cells from a grid (ICDAR 2023)

Formerge splits a table into a grid of cells via separator detection and then uses a transformer to recover spanning cells from that grid. Operationally it is a merging operator: it reduces the cardinality of a complete grid into spans, whereas the proposed insertion operator increases the cardinality of an incomplete set. The reason it matters here is design vocabulary: Formerge demonstrates that learned reasoning about which regions belong to a common spanning cell is feasible and publishable, which directly supports the plausibility of the span-compatible outcome. The paper should cite it as the span-side counterpart and explain that span-compatibility in the insertion setting means abstaining from subdividing a region that an existing span already explains.

5.5 GrabTab - flexible component deliberation (AAAI 2024)

GrabTab treats complex table recognition as progressive deliberation over flexible components, selecting reasonable structural components rather than forcing a fixed representation. It is cited here because it normalizes the idea that TSR benefits from explicit decision-making about structure, not just prediction. It does not insert candidates into an incomplete detection set and does not threaten the claim, but it belongs in the related work on structural decision-making, alongside TRACE and the separator-regression family (TSRFormer, SepFormer, SEMv3), which collectively show the field moving toward decomposed, decision-heavy TSR pipelines.

5.6 DQ-DETR-style TSR and TSRFormer - dynamic queries in primary detectors

A Pattern Recognition 2023 paper applies Dynamic Queries Enhanced DETR to robust table structure recognition, and TSRFormer (ACM MM 2022) uses separator-focused two-stage transformer detection. Both generate queries from image content rather than using a static query bank, which is mechanistically similar to the proposal's image-attention insertion queries. The difference is conditioning and role: these are primary detectors that produce the full candidate set from the image alone, without observing any surviving candidate set, and without a per-vacancy structural decision. The submission must cite both and state the distinction plainly: dynamic query generation is established; dynamic query generation conditioned on an incomplete candidate set, with insert/no-insert/span-compatible outcomes, is not.

5.7 DINO two-stage query selection and semi-supervised pseudo-label completion

Two mechanism relatives live outside TSR and should be cited preemptively. DINO (ICLR 2023) popularized mixed query selection, initializing decoder queries from encoder proposals - i.e., queries anchored on candidate content. Semi-supervised object detection (the Soft Teacher line and successors) completes missing pseudo-labels for unlabeled data during training, which is functionally learned completion of a retained set's missing boxes, but at training time and without any table-structural reasoning. Neither performs inference-time, structure-aware expansion of a detector's output set. Citing both converts a potential reviewer objection - is this not just two-stage DETR or SSL? - into evidence that the mechanism is sound and the task setting is new.

5.8 The VLM and image-to-sequence line

From TableFormer (CVPR 2022) through VAST (CVPR 2023), UniTable (2024, whose authors report surpassing GPT-4o and GPT-4-turbo with vision on table parsing), TabPedia (NeurIPS 2024), UniTabNet (EMNLP 2024), InstructTable (2026), and StrucTab (ECCV 2026), autoregressive models emit structure tokens and coordinates in one stream. In a loose sense every emitted cell is an insertion, but there is no explicit detection-conditioned insertion stage: the candidate set is not an input, no-insertion is not an outcome, and span-compatibility is implicit in HTML attributes. Cell box precision from sequence decoders also typically trails dedicated detection heads, which is precisely why a detection-level recovery module remains relevant. StrucTab deserves a sentence of its own: with intermediate structural supervision, decomposed RL rewards, and a new benchmark, it represents the strongest current VLM-side claim, and the submission should compare against or at least discuss it.

5.9 The residual gap the paper may claim

> *No published TSR system learns a variable-cardinality insertion operator that expands an incomplete, detector-produced candidate set using raster image evidence and surviving-cell context, with an explicit learned decision among insertion, abstention, and span-compatibility. Geometric completion, vacancy searching, span merging, dynamic queries, and training-time label completion each exist in isolation; their detection-conditioned, decision-aware composition does not.*
>
> *Residual gap statement supported by this audit - re-verify immediately before submission*

WHAT REVIEWERS WILL ATTACK

6. Threat Analysis and Positioning

6.1 Threat 1 - is this not just LGPMA or the IJDAR reconstruction?

This is the most likely first-round objection, and the answer is already half-built into the proposal's novelty table. The differentiation rests on four properties the priors lack: the operator is learned end to end; it is conditioned on an external detector's surviving candidate set; its output cardinality varies; and it can explicitly abstain or defer to an existing span. The submission should make this a single sharp paragraph plus a comparison table, and then let the matched-precision experiment against a reimplementation of alignment-based completion carry the argument empirically. Anything asserted in prose but not measured against that baseline will be discounted.

6.2 Threat 2 - lower the threshold or improve the detector instead

Detector-side improvement remains a live alternative: ConRTF at ICDAR 2026 gains up to +1.6 GriTS over real-time detector baselines purely through training-time edge weighting, and threshold relaxation requires no new components at all. The proposal already schedules the lower-threshold baseline; the submission should also report a candidate-budget-matched comparison (same number of retained candidates for both methods) and discuss the failure signature of thresholds - duplicates and competing false candidates rather than clean recovery. If learned insertion only wins where thresholds fail, say so explicitly; that is a stronger, more honest claim than a blanket one.

6.3 Threat 3 - why not just use a vision-language model?

The VLM wave is the biggest contextual change since the proposal was drafted: UniTable reports beating GPT-4o on table parsing, StrucTab is an ECCV 2026 system with decomposed RL rewards, and InstructTable shows instruction-guided multi-stage TSR. The submission needs a positioning paragraph that concedes the trend and then draws the line: the proposed module operates at the detection level where precise cell geometry is decided; sequence models are optimized for structure and content tokens and their boxes typically trail dedicated detectors; the recovery module is orthogonal and attachable to any detector, including one inside a VLM pipeline; and it introduces no generative content beyond verified image-grounded boxes. If compute allows, one sequence-model comparison on PubTables-1M cell localization, or a cited number from prior work, closes this thread decisively.

6.4 Threat 4 - synthetic deletion is not detector failure

The proposal handles this correctly by making natural omissions the deciding evaluation, with synthetic deletion only for controlled analysis. Two additions will harden it: first, report the omission taxonomy of the real detector (how many omissions are filtered versus proposal omissions, and their size, position, and span-adjacency distribution) so reviewers can judge ecological validity; second, include the intact-table

no-change accuracy so the abstention story is tested on real, unmodified outputs rather than only on constructed negatives.

6.5 Citation additions the paper should make

Add	Venue	Why
DINO (Zhang et al.)	ICLR 2023	Mixed query selection - queries from encoder proposals; mechanism relative that must be acknowledged.
Soft Teacher (Xu et al.) and SSL-OD successors	CVPR 2021+	Training-time pseudo-label completion of missed boxes; the inference-time contrast sharpens the claim.
DQ-DETR-based TSR	Pattern Recognition 2023	Dynamic queries for table cells in a primary detector; closest in-TSR query mechanism.
Formerge	ICDAR 2023	Learned span recovery by merging; the span-compatible outcome's counterpart.
GrabTab	AAAI 2024	Structural decision-making / component deliberation context.
SepFormer, TABLET, DTSM, SEMv3	ICDAR/IJCAI 2024-2025	Current non-VLM TSR landscape; shows the venue's expectations.
UniTable, TabPedia, UniTabNet, InstructTable, StrucTab	2024-2026	The VLM line that Threat 3 raises; cite and position against it.
PubTables-v2 (Smock et al.)	arXiv 2025	Verified real; ideal future-work testbed for full-page and multi-page settings.

Table 6.1 - Recommended citation additions beyond the proposal's current 28 references.

6.6 Recommended positioning statement

> *Rather than proposing another table parser, we study a missing operator in detection-based TSR: a learned, detector-conditioned insertion stage that expands an incomplete candidate set only when image evidence and surviving-cell context justify it, abstains when they do not, and defers to existing spanning structures. We measure it against threshold relaxation and geometric completion at matched precision, and we report separately the omissions that no candidate pool ever contained - the cases that only insertion can recover.*
>
> *Suggested positioning paragraph for the submission*

FROM AUDIT TO ACTION

7. Publishability Verdict and ICDAR 2027 Plan

AUG 18-22
ICDAR 2027 dates - Kuala Lumpur, Malaysia

28 FEB 2027
Listed paper submission deadline (verify on icdar2027.org)

15 MAY 2027
Listed acceptance notification date

Verdict. The novelty audit removes the main reason a venue would reject this work at the screening stage: the core mechanism is not in the literature, the closest priors are identified and distinguishable, and the reference list is real. What remains is ordinary scientific risk - the experiments must show what the hypotheses promise. For a solo student with one consumer GPU, this is a viable ICDAR-sized project precisely because the proposal keeps the host detector frozen and the recovery module small; the decisive experiments are inference-plus-light-training workloads, not large-scale retraining.

7.1 Two realistic paths

Path	Scope	Target	Trade-off
A - compressed conference push (recommended)	Phases 1-5 of the proposal: TATR host, controlled-deletion training, geometric and threshold baselines, image-only vs full model ablation, natural-omission evaluation on PubTables-1M. SciTSR transfer only if time remains.	ICDAR 2027 (deadline 28 Feb 2027, about 4.3 months away)	Tight but feasible; cross-dataset and second-detector studies move to the journal extension.
B - full development cycle	All seven proposal phases including WTW, SciTSR-COMP, a second host detector, and efficiency analysis.	IJDAR (rolling) or Pattern Recognition; ICDAR 2028 if preferred	Uses the full six months; a stronger journal-depth paper; delays the first publication.

Table 7.1 - Two paths consistent with the audit's findings and the stated constraints.

The recommended path is A with a journal extension planned from the start: ICDAR values document-specific insight, careful benchmarking, and honest analysis, and the intervention study plus the new evaluation protocol fits that profile. The six-month plan the user stated still works - it simply ends in a journal submission for the extended version while the conference version is prepared inside the first four and a half months.

7.2 Feasibility notes for one consumer GPU

- Freeze the host: TATR structure-recognition weights are public (Hugging Face, MIT license) and require no retraining; run inference once over the training split and cache candidate sets and image features so the recovery module trains from caches.

- Keep the module small: a ResNet-50/FPN or even a frozen TATR backbone feature pyramid plus a set-transformer context encoder and a light query decoder stays in the few million-parameter range and trains on a single 24 GB card.
- Subsample PubTables-1M: 100-200K tables are enough to train and validate the module; report the subset policy. Synthetic deletion generation is CPU-side and cheap.
- Budget the eval: natural-omission evaluation is one forward pass per test table plus matching bookkeeping; plan storage for per-table predictions to enable the filtered-versus-proposal analysis.

7.3 The killer experiment list

#	Experiment	Decides
E1	Natural omission profile of TATR on PubTables-1M test at the operating threshold, split into filtered vs proposal omissions	H1; whether the problem is big enough to matter
E2	Lower-threshold baseline at matched candidate budget	H2; the cheapest competitor
E3	Geometric completion baseline (alignment-based, reimplemented on the same detector)	Whether learned insertion beats the closest prior mechanism
E4	2x2 ablation: context-only, image-only, full model, no-recovery control	H4; both evidence sources matter
E5	With-vs-without no-insertion class and span-compatible head	H5; span violation rate and no-change accuracy
E6	Headline: recovery recall on proposal omissions at matched insertion precision (IoU 0.50 and 0.75)	The paper's central empirical claim
E7	End-to-end GriTS/TEDS with the recovery module attached	Whether the intervention improves the final structure metric, reported after E1-E6

Table 7.2 - Minimum evidence set for a credible ICDAR submission.

The proposal's own go/no-go rule remains the right one and is endorsed unchanged: proceed to the larger architecture only if the full model beats geometric completion and image-only recovery on natural omissions at matched insertion precision, with gains that include proposal omissions absent from the generous pre-filter pool. If the gate fails, the honest paper is an analysis of where and why learned insertion does not beat rule-based completion on canonicalized data - ICDAR publishes such studies, and the evaluation protocol alone keeps it useful.

7.4 Suggested framing

Keep the proposal's title direction; it is accurate and specific. One refinement for the conference version: foreground the decision, not the architecture, because the three-way decision is the most defensible novelty. A candidate title is: **When to Insert: Learning Detection-Conditioned Cell Recovery for Table Structure Recognition**, with the abstract leading on the filtered-versus-proposal omission distinction and

the matched-precision comparison. The main figure should be the proposal's Figure 1 extended with one hard negative - a vacancy that the model correctly refuses to fill because a spanning cell explains it; reviewers remember the refusal case longer than the recovery case.

ANNOTATED AND LINK-CHECKED OCTOBER 2026

8. Verified Reading List

The following items were located and verified during this audit. Group 1 is the host platform and evaluation context; Group 2 is the closest prior art the paper must be written against; Group 3 is the mechanism-relevant adjacent work; Group 4 is the 2025-2026 TSR landscape the related-work section must reflect; Group 5 is metrics and citation hygiene.

Group 1 - host detector, datasets, evaluation context

B. Smock, R. Pesala, R. Abraham. PubTables-1M: Towards Comprehensive Table Extraction from Unstructured Documents. CVPR 2022. (arXiv:2110.00061; TATR weights on Hugging Face, MIT license.)

B. Smock et al. PubTables-v2: A New Large-Scale Dataset for Full-Page and Multi-Page Table Extraction. arXiv:2512.10888, 2025. First multi-page TSR benchmark; verified during this audit; ideal future-work testbed.

Z. Chi et al. SciTSR: A Dataset for Complex Scientific Table Structure Recognition. 2021. 15,000 scientific tables; SciTSR-COMP complex subset.

R. Long et al. Parsing Table Structures in the Wild. ICCV 2021. WTW; deformation stress test.

Group 2 - closest prior art (write the paper against these)

L. Qiao et al. LGPMA: Complicated Table Structure Recognition with Local and Global Pyramid Mask Alignment. ICDAR 2021, pp. 99-114. Empty-cell searching and merging; ICDAR 2021 best industry paper.

L. G. Singh, S. Middleton. Tabular Context-Aware Optical Character Recognition and Tabular Data Reconstruction for Historical Records. IJDAR, vol. 28, 2025. (s10032-025-00543-9) Alignment-based missing-cell inference - the geometric completion baseline.

L. G. Singh et al. Data Rescue of Historical Tables through Semi-Supervised Table Structure Recognition. IJDAR, 2025. (s10032-025-00562-6)

N. Q. Nguyen et al. Formerge: Recover Spanning Cells in Complex Table Structure Using Transformer Network. ICDAR 2023, pp. 522-534.

H. Liu et al. Grab What You Need: Rethinking Complex Table Structure Recognition with Flexible Components Deliberation. AAAI 2024.

Robust Table Structure Recognition with Dynamic Queries Enhanced Detection Transformer. Pattern Recognition, 2023. (DQ-DETR-based TSR)

Group 3 - mechanism relatives outside TSR (cite preemptively)

H. Zhang et al. DINO: DETR with Improved DeNoising Anchor Boxes for End-to-End Object Detection. ICLR 2023. (arXiv:2203.03605; mixed query selection.)

M. Xu et al. End-to-End Semi-Supervised Object Detection with Soft Teacher. CVPR 2021. Training-time pseudo-label completion line.

Group 4 - 2025-2026 TSR landscape (verified during this audit)

E. Thomas et al. ConRTF: Edge-Constrained Boundary Distribution Refinement for Realtime TransFormer Table Structure Recognition. ICDAR 2026. (arXiv:2607.00734)

L. Hamdi et al. TableSeq: Unified Generation of Structure, Content, and Layout. arXiv:2604.16070, 2026.

L. Hamdi et al. FastTab: A Fast Table Recognizer with a Tiny Recursive Module and 1D Transformers. arXiv:2605.22422, 2026.

G. Li et al. StrucTab: A Structured Optimization Framework for Table Parsing. ECCV 2026. (arXiv:2606.29905; TableVerse-5K benchmark.)

J. Li et al. Text Prior-Guided Cell Alignment with Transformer for Robust Table Structure Recognition. Pattern Recognition, 2026, 114832.

L. Liu, C. Yang. OG-HFYOLO: Orientation Gradient Guidance and Heterogeneous Feature Fusion for Deformation Table Cell Instance Segmentation. arXiv:2504.20682; includes the DWTAL dataset.

A. Xiao, C. Yang. Towards One-Stage End-to-End Table Structure Recognition with Parallel Regression for Diverse Scenarios (TableCenterNet). arXiv:2504.17522, 2025.

L. Hu, S. Huang. Enhancing Table Structure Recognition via Bounding Box Guidance (BGTR). ICPR 2024. (arXiv:2609.08705, archival upload Sep 2026 - cite as ICPR 2024.)

L. Hamdi et al. InstructTable: Improving Table Structure Recognition Through Instructions. arXiv:2604.02880, 2026.

S. Peng et al. UniTable: Towards a Unified Framework for Table Recognition via Self-Supervised Pretraining. arXiv:2403.04822, 2024. Reports surpassing GPT-4o on table parsing; the VLM-positioning anchor.

SepFormer: Coarse-to-Fine Separator Regression Network for Table Structure Recognition. ICDAR 2025. TABLET: Table Structure Recognition using Encoder-only Transformers. ICDAR 2025. DTSM. ICDAR 2024. SEMv3. IJCAI 2024.

Group 5 - metrics and citation hygiene

B. Smock, R. Pesala, R. Abraham. GriTS: Grid Table Similarity Metric for Table Structure Recognition. WACV 2023. (arXiv:2203.12555.)

X. Zhong, J. Tang, E. Yepes. Image-based Table Recognition: Data, Model, and Evaluation. ECCV 2020. (arXiv:1905.13391.) This is the correct TEDS citation - replace the proposal's reference [19] (arXiv:2208.00385).